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Unit – 3 : Session – 8 : SLO – 1 | Queue — Introduction

■ I. Fill in the Blanks

- A queue follows the FIFO (First-In-First-Out) principle.
- In a queue, insertion is performed at the REAR end.
- In a queue, deletion is performed at the FRONT end.
- A SIMPLE (linear) queue is a linear data structure with a fixed size.
- A CIRCULAR queue overcomes the limitation of fixed size by wrapping around.
- A DEQUE (double-ended queue) allows insertion and deletion from both ends.
- PRIORITY queue gives priority to elements based on certain criteria.
- Queues are used in CPU (process/job) scheduling in operating systems.
- In a queue, if the rear is just behind the front in a circular manner, the queue is said to be FULL.
- An ARRAY (or linked list) is used to implement queues in programming languages like C.
- The queue operation that checks whether it is full is called ISFULL.
- The queue operation that checks whether it is empty is called ISEMPY.
- In breadth-first search (BFS) of a graph, A QUEUE is used to keep track of the next node to visit.
- In printers, tasks are managed using a QUEUE to print in the order they arrive.
- A queue where new elements are added based on priority rather than position is called a PRIORITY QUEUE.